MATPLOTLIB (DAY-1)

- Data visualization is an essential skill for all data analysts and Matplotlib is one of the most popular libraries for creating visualizations.
- `matplotlib` is a Python package used for data plotting and visualisation. It is a useful complement to Pandas, and like Pandas, is a very feature-rich library which can produce a large variety of plots, charts, maps, and other visualisations.

PLOT

1. **plot() - It is used to plot a 2-D graph**

   Syntax-
   plt.plot(x,y)

   number of values in x == number of values in y

2. **title() -It is used to give title to plot**
   Syntax-
   plt.title('title')

3. **xlabel()- It is used to label the x-axis**
   Syntax-
   plt.xlabel('label')

4. **ylabel()- It is used to label the y-axis**
   Syntax-
   plt.ylabel('label')

5. **legend() - It function is used to label the elements plotted on the graph**
   Syntax-
   plt.legend()

   In order add a legend it is necessary provide a `label` argument to your plotting functions (like `plt.plot()` or `plt.scatter()`)

6. **grid()- It is used to major grid lines on both the x and y axes by default.**
   Syntax-
   plt.grid()

7. **show()- It is used to display only the graph**

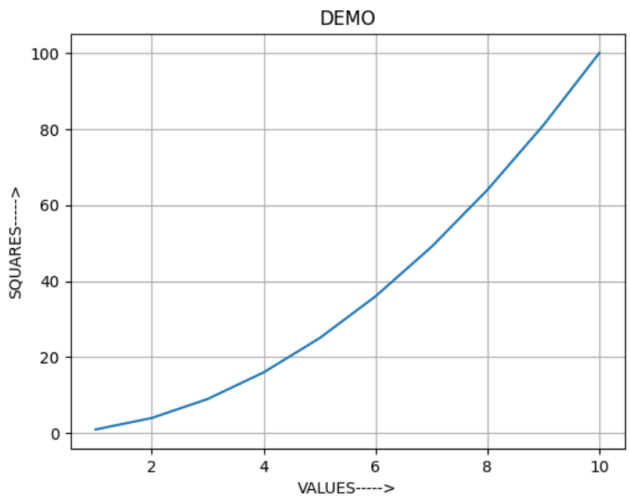
   Syntax:
   plt.show()

8. **OPTIONAL**
   xticks() ,yticks()-It is used to control the tick marks and their labels on the x and y axes respectively.

   Syntax:
   plt.xticks()
   plt.yticks()

```
a=np.arange(1,10+1)
b=a**2

plt.plot(a,b)
plt.title('DEMO')
plt.xlabel('VALUES----->')
plt.ylabel('SQUARES----->')
plt.grid(True)
plt.show()
```



ATTRIBUTES

▼ **Attributes of plot()**

1. color- It is used to change the color of the plotted line
2. label- It is used to label the plotted line
3. linewidth- It is used to change the width of the line
4. linestyle- It is used style the plotted line
   Types of linestyle are-
   '-', '--', '-.', ':', 'None', ' ', '', 'solid', 'dashed', 'dashdot', 'dotted'
5. marker- It is used to mark the plotted points
   Types of marker are-
   'o','^','v','.'
6. markersize- It is used to change the size of marker
7. markeredgecolor - It is used to give color for the marker outline
8. markerfacecolor- It is used to give color for the marker

▼ **Attributes of xlabel() and ylabel() and title()**

1. color- It is used to change the color of the labels
2. fontstyle- It is used to change the fontstyle of the label
   Types of fontstyle are - 'normal', 'italic', 'oblique'
3. fontsize- It is used to change the size of the font

▼ **Attributes of legend()**

1. loc- It determines the place where legend is placed
   Types of loc are- 'best', 'upper right', 'upper left', 'lower left', 'lower right', 'right', 'center left', 'center right', 'lower center', 'upper center', 'center'
2. title- It is used to give title to legend
3. facecolor- It is used to give color to the legend
4. labelcolor-It is used to change the color of the text labels

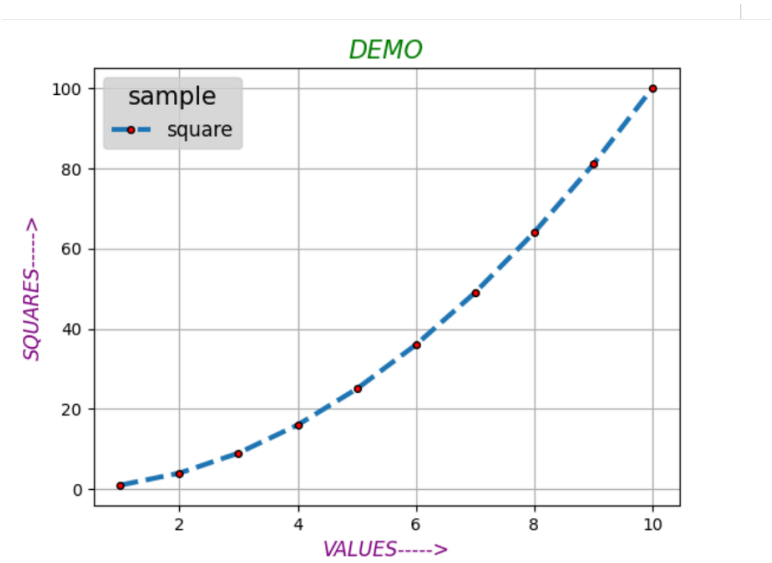
5. fontsize- It is used to set size of the font in legend except title of the legend
6. title_fontsize- It is used to set size of the font of the title

▼ **Attributes of xticks() and yticks()**

We usually use this only if we want to modify the by default values of the ticks

1. labels- It is used to change the labels of the axis
2. fontsize-It is used to change the font size of the tick labels
3. color-It is used to change the color of the tick labels
4. rotation -It is used to adjust the angle of the tick labels
   It is usually recommended to be kept 45 if the labels with each other

```
plt.plot(a,b,label='square',linestyle='dashed',linewidth=3,
marker='o',markersize=4,markerfacecolor='red',markeredgecol
or='black')
plt.title('DEMO',color='green',fontsize=15,fontstyle='itali
c')
plt.xlabel('VALUES----->',color='purple',fontsize=12,fontst
yle='italic')
plt.ylabel('SQUARES----->',color='purple',fontsize=12,fonts
tyle='italic')
plt.legend(loc='best',facecolor='lightgray',title='sample',
fontsize=12,title_fontsize=15)
plt.grid(True)
plt.show()
```



Annotate

- **annotate()-It is used in highlight the data points of the plot**
- **Case-1**
  It is used to label all the co-ordinates of the plot

```
for i,j in zip(x,y):
  plt.annotate(f'{i,j}',(i,j))
```

plotting the data point(what to write)

particular position (where to write)

Case-2
To highlight specific co-ordinates of the plot

Syntax:
plt.annotate(xy,text)
where ,
xy⟶Where to label
text⟶What we want to label

fontsize- It is used to give size to the text

color- It is used to give color to the text

ha- It is used to adjust the horizontal alignment of the label
Types of ha are-'center','left',right'

ADDITIONAL FUNCTIONS

- **figure()- It creates a blank white sheet on which you will draw your plot by default**
  Python automatically creates a default canvas for you. You only use it when you want to control the size or style of your plot.

  figsize- It is use to adjust the width & height in inches .figsize= (width,height)

  facecolor- It is used to change the background color of the figure

  edgecolor- It is used to change the color of the border of the figure

- **savefig()- Exports the current Matplotlib figure from memory to a file**

  Syntax:
  plt.savefig('plot')

  plt.savefig ('plot.png')⟶PNG FORMAT
  plt.savefig ('plot.jpg')⟶JPEG FORMAT
  plt.savefig ('plot.pdf')⟶PDF FORMAT

  Always call `plt.savefig()` **before** `plt.show()`

- **axhline() and axvline()- It is used to draw a horizontal or vertical line across the entire plot at a specified value.**

  Syntax:
  plt.axhline(value)
  plt.axvline(value)

  ATTRIBUTES-

color- It is used to change the color of the line

linestyle- It is used to change the style of the line

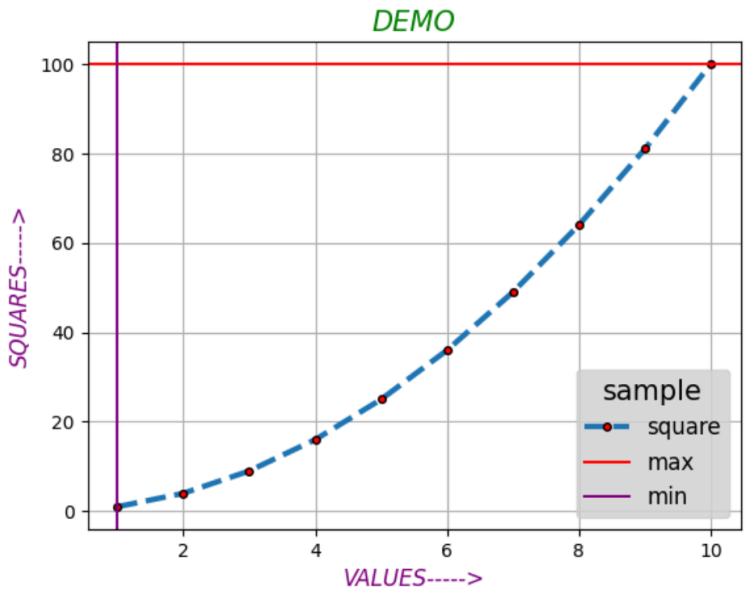
linewidth-It is used to change the width of the line

label- It is used to give label to the line. Using this attribute the line will get highlighted in the legend too

Important - It is necesaary to use axhline() and axvline() before plt.legend()

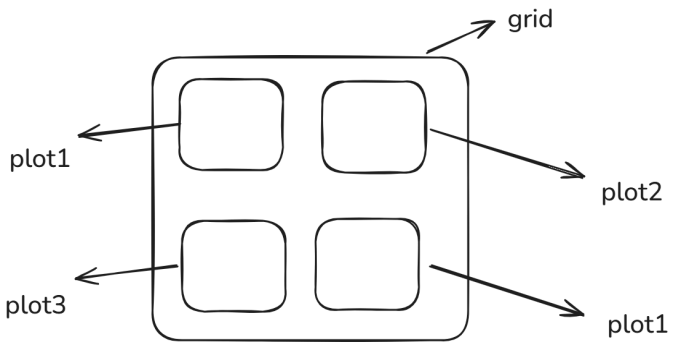
```
plt.plot(a,b,label='square',linestyle='dashed',linewidth=3,
marker='o',markersize=4,markerfacecolor='red',markeredgecol
or='black')
plt.title('DEMO',color='green',fontsize=15,fontstyle='itali
c')
plt.xlabel('VALUES----->',color='purple',fontsize=12,fontst
yle='italic')
plt.ylabel('SQUARES----->',color='purple',fontsize=12,fonts
tyle='italic')
plt.axhline(b.max(),color='red',label='max')
plt.axvline(a.min(),color='purple',label='min')
plt.legend(loc='best',facecolor='lightgray',title='sample',
fontsize=12,title_fontsize=15)
plt.grid(True)

plt.show()
```



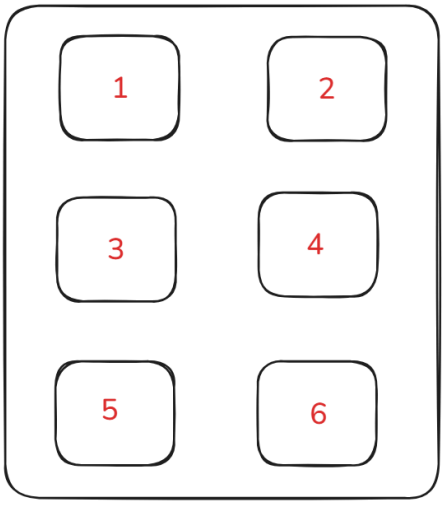
SUBPLOT

- It is used to plot multiple plots in a single grid



- Assume grid like a entire chocolate bar and the individual chocolate pieces of the chocolate bar are the plots
  Syntax:
  plt.subplot(rows,cols,position)
- It will create imaginary grids where you can place a plot in any position of that grid whenever you use it
- **Example -**

  plt.subplot(3,2,1)



- **plt.subplot(3,2,1)→only 2nd grid will be visible rest of them will be hidden**
- **plt.subplot(3,2,7)→ERROR SINCE THERE ARE NO ENOUGH PLOTS**
- **tight_layout()-It is used to adjust the spacing between subplots so that they don't overlap each other.**

  Syntax:
  plt.tight_layout()